UGANDA MARTYRS S.S. NAMUGONGO

" perseverance and success."

GRAND A-LEVEL MATHEMATICS SEMINAR 2026

DISCUSSION ITEMS BOOKLET

Date: 3rd October 2026

PRINCIPAL MATHEMATICS PAPER 1

CONSTRUCT 1: GEOMETRY

ITEM 1

A businessman is setting up a large commercial distribution hub in downtown Kampala to manage his matooke truck delivery routes, organize three local dairy retail shops, and install parallel underground water pipes within the facility's concrete foundation grid.

The boundary path taken by the delivery truck forms a smooth curved loop around Kampala city such that the ratio of its distance from a key market depot at position $F(6, 2)$ to its distance from a main highway security line given by $x = \frac{34}{3}$ is $3 : 5$. The businessman needs to know the clear grid equation for this delivery truck loop path and state the central focal points of the path. A police patrol car drives along a straight road modeled by the equation $mx + 4y = 25$ that just touches the delivery car path, where m is a constant parameter that determines the sloping of the path . The patrol team states that the path is safe if m is less than 2 and non negative otherwise its unsafe.

Directly under the truck route, he sets up three local dairy distribution shops that form a triangle shape marked ABC. The corner angle θ at shop C The angle θ at shop C is determined by two dimensionless design factors x and y, using the rule $\theta = \tan^{-1}(2x) + \tan^{-1}(3y)$. He needs to know the expression for $\tan\theta$ in terms of the factors x and y. On a busy Saturday, the factors are tightly locked under a working values $x = \frac{1}{2}$ and $y = 1$. With these factors locked, the actual distance between shop A and shop C is measured as 40 meters, and from shop B to shop C is 45 meters. The businessman needs to know the distance from shop A to shop B, and total floor space of the shop zone. A local builder counts the total steps around the whole layout and estimates the total to be roughly 160 meters but he's not sure if the measurements are okay.

Beneath the floor of the dairy shops, two underground water pipelines run along straight, non-intersecting paths through a concrete support block. These paths have vector lines $L_1 : \mathbf{r} = \begin{pmatrix}1\\2\\2\end{pmatrix} + \lambda\begin{pmatrix}2\\-1\\0\end{pmatrix}$ and $L_2 : \mathbf{r} = \begin{pmatrix}3\\3\\0\end{pmatrix} + \mu\begin{pmatrix}-4\\2\\0\end{pmatrix}$. The Businessman needs to be sure if some point $P(3, 2, 2)$ lies on the flat surface of the concrete foundation plane Π that holds both water pipelines together.

Task

Using appropriate mathematical calculations, help the businessman determine the required quantities and make valid decisions about the proposed developments.

ITEM 2

A town council of Kira town is planning to improve the road network of the town. As part of the first phase, the main road runs along a straight path represented by the equation $L_1 : 2x = 3y - 6$. The managing committee of Kira Market, located at point $P(8, 2)$ requires a short access road connecting the market directly to the main road. To minimize construction costs, the council intends to construct the shortest possible straight access road from the market to L_1. However, they only have enough materials to construct 5 units of road and must determine if this supply will be sufficient. Additionally, the council plans to construct another connecting road to the community hospital, represented by $L_2 : x + 4y = 12$. For safety, the acute angle between the main road L_1 and the proposed connecting road L_2 must be greater than $45°$ to ensure a safe junction. The council must therefore evaluate whether this proposed layout satisfies both the material constraint and safety criteria.

Building upon these road network upgrades; the town council is also installing a remotely operated road barrier at one of the new junctions. To automate this, the council designed a computer system that controls the raising and lowering of the barrier. The movement of the barrier is modeled by the function:

$$f(\theta) = 5\cos 2\theta + \sqrt{3}\sin 2\theta - 2\sin\left(2\theta + \frac{\pi}{2}\right) + \frac{8\sqrt{3}}{5}$$

where θ represents the angle through which the barrier rotates. The computer system will only be approved for deployment if at least two values of θ, measured in radians satisfy $f(\theta) = 0$ for $0 \leq \theta \leq \pi$.

Finally, to further improve traffic flow in the community, the town council proposes constructing two additional routes, Road A and Road B. The paths of the two roads are represented by:

- **Road A:** $\mathbf{r} = (\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}) + t(2\mathbf{i} - \mathbf{j} + 3\mathbf{k})$
- **Road B:** $\dfrac{x-3}{4} = \dfrac{1+y}{2} = \dfrac{8-z}{1}$

For this final phase of the road network to be approved, both roads must intersect precisely at the community's main bus stop, located at $B(7, 1, 7)$. Furthermore, the acute angle between the two roads must be greater than $78°$ to allow vehicles and pedestrians to navigate the junction safely. The council must determine whether these proposed roads satisfy both the structural and safety requirements.

Task

With clear mathematical evidence, assist the town council in making informed decisions regarding all three project proposals.

ITEM 3

A modern commercial eco-farm in Wakiso is setting up its automated infrastructure. The technical team needs to finalize layouts for their solar energy grid, communications tower, and automated drone delivery paths. All coordinates are measured in meters. The farm is installing a curved mirror trough to focus sunlight onto a central heat-collecting pipe. On a planned coordinate grid system, a point $P(x, y)$ on the mirror's surface moves such that its distance from a vertical boundary fence line ($x = -3$) is always exactly equal to its distance from a fixed grounding pin located at $F(3, 1)$. The installation is successful only if the length of the line passing through the focus of the mirror from one end to the other but perpendicular

to the axis of symmetry of the mirror is beyond $10m$

The farm engineer wants to measure the height of one of the buildings on the farm so as to install a lightning conductor. He uses an instrument that can measure the angle between two points and makes use of the nearby communication tower of height $20m$. While on top of the building, he views the top and bottom of the tower, and notes that the combined angle between his two sight-lines is $120°$. Given that the tower and the building are separated by a horizontal distance of $5m$. He will install the lightning conductor only if the height of the building is greater than $15.0m$

An automated drone delivers fertilizer across the farm. Its straight-line flight path is modeled by equation: $\mathbf{r} = \begin{pmatrix}1\\2\\5\end{pmatrix} + \lambda\begin{pmatrix}2\\-1\\4\end{pmatrix}$ The flying drone may operate safely only if the distance between its path and the plane modeled by the equation: $\mathbf{r}\cdot\begin{pmatrix}1\\2\\0\end{pmatrix} = 11$ is preferably be greater than $3m$

Task

Using appropriate mathematical calculations, help the farm make valid decisions regarding the three proposed developments.

ITEM 4

A growing community is designing a new public park area.

The community wants to build a circular local market. A surveyor tracks a moving point $P(x, y)$ that moves such that its distance from point $A(3, 5)$ is twice as far as it is from point $Q(-2, 3)$. A straight main supply road is planned to run along the path given by the line equation $3x + 4y = 25$.The main road must just touch the boundary of the local market, otherwise the moving point stands to be revised.

The community also has a green triangular garden plot ABC is designed with side lengths $a = 9$ meters, $b = 6$ meters, and $c = 10$ meters. To keep the plants watered, a circular sprinkler system is installed perfectly around the three corners of the triangle. Additionally, a decorative roof like structure is built above the garden, where its height h at any point changes according to the formula: $h = 2\cos^2 x + \sqrt{3}\sin 2x + 5$ where x is the angle of inclination of the support stands. Local environment rules state that the extra sitting area between the edge of the triangle and the outer watering circle must be greater than 50 m^2, and the highest point of the roof like structure must be less than 9 meters.

Three security flying drones patrol the park sky. To prevent accidents, engineers must study their straight-line flight paths, which are given by the following three line equations: $L_1 : \mathbf{r} = \begin{pmatrix}2\\3\\2\end{pmatrix} + t\begin{pmatrix}1\\1\\2\end{pmatrix}$, $L_2 : \mathbf{r} = (6 + 2\lambda)\mathbf{i} + (6 + \lambda)\mathbf{j} + (7 + \lambda)\mathbf{k}$, $L_3 : x - 2 = \frac{1-y}{-2} = z - 4$. The paths are safe only if all three paths are concurrent at some point and the angle between L_2 and L_3 is greater than $30°$.

Task

Basing on the mathematical evidence, help the community make valid decisions.

ITEM 5

A district planning team is working on three different technical project. A team of engineers is mapping out a triangular community plot on a coordinate grid to install a new network tower. The three corners of the plot are located at grid positions $A(6,10)$, $B(2,2)$, and $C(14,2)$. The engineers want to place the heavy telecom mast at the point where the three altitudes of the triangle meet and they also want to place the power generator at a separate point that is exactly the same distance from all three corners. The placement is allowed only if distance between the mast and the generator is greater than $2.5m$.

An automated greenhouse uses a mechanical arm to open its roof windows to keep the crops cool. The angle x at which the window opens must satisfy the following wave equation: $\sin 3x + \sin x + \sin 2x = 0$.To prevent strong winds from breaking the window frame, the glass panels can only be opened safely through an acute angle; otherwise, the window may be damaged

A new storage warehouse is being built with a modern V-shaped roof consisting of two flat, sloping surfaces. On a 3D design grid, the equations of the two roof planes are given by:$x + 2y - z = 2$ and $2x - y + z = 3$. For structural stability, he main ridge line where the two roof planes meet must pass through a concrete support pillar located at $(2,-1,-2)$. Additionally, to ensure heavy rainwater drains off quickly, the angle between the two sloping roofs must be greater than $75°$.

Task:

Using Mathematical justification, help the district planning team to make valid decisions.

ITEM 6

You are a Lead Structural Analyst working on a modern, pyramid-shaped pavilion project. To receive a formal building permit and be certified as operational, your design must satisfy three distinct spatial, environmental, and mechanical constraints. The base of the pyramidal pavilion follows a flat plane mapped by the equation:$\vec{r} = \begin{pmatrix}1\\2\\3\end{pmatrix} + \lambda\begin{pmatrix}2\\3\\4\end{pmatrix} + \mu\begin{pmatrix}-3\\8\\5\end{pmatrix}$. The apex vertex of the pyramid structure is located at the coordinate $V(4,-5,12)$. The pavilion will be approved only if its perpendicular height from the apex to the base plane is greater than 7.5 units.

An automated drainage trench must be channeled to protect the foundation from heavy rainwater runoff. Geotechnical benchmarks are fixed at points $A(-2,6)$ and $B(4,-8)$. To balance water pressures symmetrically, the trench must run along the perpendicular bisector of the line segment AB and that the angle between the trench and another line $2x-4y = 5$ must be greater than $4.8°$ otherwise the pressure becomes unbalanced.

An adjustable solar canopy attached to the pavilion has a tilt factor θ that must satisfy the equation:$3\sin(2\theta) + 4\cos(2\theta) = 2$ The tilt is safe only if $120° \leq \theta \leq 240°$

Task

Using appropriate mathematical calculations, make valid decisions regarding the three proposed systems.

CONSTRUCT 2: ALGEBRA.

ITEM 7

A company is testing an autonomous drone-delivery system for a large agricultural estate. The drone takes off from a central control station O, represented by the origin on an Argand diagram. During one test flight, the drone's possible horizontal distances x from the control station are modelled by $x^2 - 2(k+3)x + k^2 + 4k + 13 = 0$, where k is a parameter associated with the drone's flight setting. The company wants to determine the condition on k for the two possible horizontal distances to be distinct real values. The company will approve the flight setting only when this condition is satisfied.

During another flight, the drone's position relative to the control station is represented by the complex number $Z = -4 + 4\sqrt{3}\,i$. The engineers want to represent the drone's position using distance and direction rather than Cartesian coordinates.

The drone is programmed to pass through four checkpoints represented by the solutions of $w^4 = 16$.. The drone can safely use these checkpoints only if one of the complex coordinates is negative. The company has specified that a checkpoint is within the safe horizontal operating interval if its real coordinate x also satisfies $-1 \leq x \leq 3$.

Task

Using appropriate mathematical calculations, help the company determine the required quantities and make valid decisions about the proposed developments

ITEM 8

A coffee-processing cooperative in Masaka District is preparing for the new harvesting season. As part of its preparations, the cooperative is improving its marketing facilities, checking equipment used in handling coffee and preparing land for smallholder coffee farmers.

The cooperative is establishing a square-shaped demonstration garden within its premises. The garden will be used to demonstrate improved agricultural practices to smallholder coffee farmers. The area of the garden is $18 + 12\sqrt{2}$ cm^2. Before beginning the work, the cooperative needs to completely enclose the garden. It has 24 cm of fencing material available for this purpose. The cooperative wants to establish whether the available fencing material will be sufficient to cover the entire perimeter of the garden.

The cooperative also uses a digital weighing scale to determine the quantities of coffee delivered by farmers. Before the scale is put into operation, its calibration system is tested using two settings

represented by x and y. The calibration report shows that one reading is $2\log_y x$ and the other is $2\log_x y$. For the calibration to be accepted, the sum of the two readings must be 5. The report also records that the product of the two settings is 27. The manufacturer's requirements state that the two settings must be positive whole numbers, with the larger setting being three times the smaller setting, before the weighing scale can be put into operation.

At the same time, the cooperative has received $50,100\ \text{m}^2$ of land for establishing plots for smallholder coffee farmers. To ensure fairness, the areas of the plots are to increase in an arithmetic progression. The smallest plot is $2\ \text{m}^2$ and the largest is $1,000\ \text{m}^2$. During the first phase of allocation, the cooperative intends to allocate the first 13 plots.

The cooperative will approve the first-phase allocation only if the total area of the first 13 plots does not exceed $900\ \text{m}^2$.

Task

Using mathematical justification, help the cooperative to:

(a) Decide whether the demonstration garden can be completely enclosed using the available fencing material.

(b) Decide whether the digital weighing scale should be put into operation.

(c) Decide whether the cooperative should approve the first-phase allocation.

ITEM 9

A certain agricultural farm produced large amounts of coffee and decided to export the coffee to a nearby country. This would require 10 weeks for the trucks to deliver the coffee. Two preservatives are added to the coffee to keep it fresh during this period. The first preservative is of concentration, $C_1 = \log_2(m+3)$. The second is of concentration $C_2 = \log_4(m+13)$ where m is the strength of the preservative. For these preservatives to work efficiently they should be mixed in equal amount and the strength must be at least 3.

During this 10 week period, senior drivers will be given a travel allowance which will increase by a constant amount every week. Driver A has been informed that in the second week she will receive shs 100,000 and in the 6th she will receive shs 180,000. He wishes to save all this money and pay tuition for his daughter of shs 1.5 million for university. But he doesn't know whether the allowance will be enough.

The farm management also gives an advance of this allowance for n weeks worked only if the square of number of weeks worked plus 40 is less than 13 times the number of weeks worked. Another driver B on the farm wishes to get an advance in the sixth week but he is not sure whether his request will be approved.

Tasks: Using your mathematical skills, help the;

a) farm management to know whether the preservative will meet their target.

b) drivers to make decisions.

ITEM 10

Tumaini Crafts Ltd., a registered youth empowerment enterprise in Mukono, exports premium eco-friendly square mats woven from banana fibers. The enterprise is expanding its operations and has hired a human resource consultant to structure their upcoming 6-month contract cycle for new weavers. The management wants to offer two distinct short-term compensation packages:

- Under Scheme A, a weaver receives Shs. 5,000,000 in the first month, but the payout is halved each subsequent month.
- Under Scheme B, a weaver starts with Shs. 250,000 in the first month, which increases by Shs. 200,000 every new month.

The Board of Directors will only approve a scheme if the total package paid to a single weaver over the 6-month period does not exceed a hard budget cap of Shs. 10,000,000. If both schemes are under budget, the scheme that maximizes worker satisfaction by providing the highest total earnings must be chosen.

The technical production team designs custom square mats whose surface area (x, in square meters) must perfectly satisfy the structural stability model: $x^2 - 16x + 36 = 0$. To minimize raw material wastage, the cutting machines require exact mathematical values. Due to the physical limitations of the weaving looms, the enterprise cannot manufacture any single mat whose side length exceeds 4.5 meters, otherwise it causes wastage of raw materials.

Financially, the enterprise must secure its international export licence by shipping maximum of large cartons of mats to overseas buyers. To meet this target, the operations manager plans to distribute the workload across N specialized worker departments. Due to the available production resources, each department is structurally optimized to pack exactly $(16 - N)$. Due to the available floor space in the Mukono factory, the facility can have a maximum of 10 active departments at any given time.

Tasks:

(a) Justify your choice by showing how the total expenditure of each scheme compares against the board's budget cap.
(b) Ascertain whether the design model yields dimensions that waste raw materials.
(c) Determine if the flow space is enough to house the departments.

ITEM 11

A youth group in Bukomansimbi District plans to establish a vegetable garden near their community centre. The expected profit y in thousands of Uganda shillings from selling vegetables depends on the number of units sold x in hundreds. The relationship between the number of units sold and the expected profit is given by $y = 5 + 12x - x^2$. The group will start the project if the maximum expected profit is at least UGX 40,000.

The group wants to connect the garden to a nearby water tank. Let x be the parameter value used to determine the lengths of the two pipes. The length of the first pipe $P_1 = \sqrt{x - 6}$ metres and the length of the second pipe $P_2 = \sqrt{2x - 11}$. metres. The two pipes together must have a total length of 5 metres. The group will buy the pipes only if x is a whole number.

The group needs to estimate the amount of water that can pass through a small section of the channel. Let x represent a small measurement used in the design. The expression used is $\sqrt{\frac{3+x}{3-x}}$. The engineers need

to test the expression by expanding it up to and including the term in x^2. For one section of the channel, $x = 0.3$. The engineer will accept the approximation if its error is less than 0.01.

Task

Using mathematical calculations, help the group make informed decisions.

ITEM 12

A secondary school in Masaka District plans to open a small school shop to provide learning materials to students.

The school buys exercise books, pens and mathematical sets as shown in the table below.

Purchase	Exercise books	Pens	Mathematical sets	Total cost
First	2	1	1	21, 000
Second	1	2	1	20, 000
Third	1	1	2	19, 000

The school will stock in large quantities only if the total cost of 5 sets, 3exercise books and 4 pens is not more than UGX 60000.

The school records the number of packets of snacks sold during a fundraising activity. The number of packets sold follows an arithmetic progression. At first it sells 81 packets. The number of packets at first, fourth time and eighth time form the first three terms of a geometric progression. The school needs to sell at least 780 packets during the five stages to meet its target.

The school uses two methods to advertise the school shop. The number of people reached using the first advertising method $C_1 = 9^x$ and the number of people reached using the second advertising method $C_2 = 3^{x-1}$,where x is the number of stages of advertising. At one stage, the difference between the numbers of people reached by the second method from the one is 22. The school will continue using the first advertising method if x is a positive value not greater than 2.

Task

Using mathematical evidence, guide the school in making proper decisions.

ITEM 13

A community group in Wakiso District is raising money to construct a community hall. The group decides to deposit UGX 400,000 at the beginning of each year into a bank account which pays compound interest at 10% per annum. The deposits are made for 5 years. The community needs at least UGX 2,600,000 from the savings account before construction of the hall can begin.

During the construction project, the committee uses a mathematical model represented by $z^3 + pz^2 + qz - 15 = 0$,where z be a measurement obtained from the construction project, while p and q are unknown

constants in the equation. It is known that $i - 2$ is one of roots of the equation. The committee will use the mathematical model only if both p and q are integers.

The community is selecting a football team for a local tournament. There are 16 players, of whom 3 are goalkeepers. An 11-players team is to be selected, and the team must contain exactly one goalkeeper.The organisers will consider the selection exercise fair only if the number of possible teams is more than 800.

Task

Basing on mathematical calculations, help the community make valid decisions.

ITEM 14

An environmental engineering firm is designing a series of stepped gravity-fed water purification pools for a green community park. The depths of the pool basins below a benchmark reference level form an arithmetic progression. The depth of the eighth pool basin is exactly twice the depth of the third pool basin. The total combined depth of the first eight basins is 39 meters. The structural engineers wish to completely redesign the pool foundation if the total combined depth of the first three pools turns out to be greater than 9 meters.

The project's chemical safety team is monitoring chemical concentration decay pathways near the primary filtration reservoir by placing two electronic tracking sensors at a distance of x (where x is measured in tens of meters) from the filtration core. One tracking sensor displays a specialized density reading of $(\log_4 x)^2$, while the second sensor indicates a reading of $2\log_2 x$.

For the bioreaction to stabilize perfectly without contaminating the local soil, the sum of these two sensor readings must equal exactly 3. The safety team intends to re-adjust the physical configuration of the sensors if their actual distance from the filtration core is found to be less than 20 meters.

Concurrently, the micro-grid electrical team is planning to install at least four low-voltage power distribution nodes on the facility's coordinate floor plan. The primary master node is positioned at the complex number coordinate $-8 + 8\sqrt{3}i$ on the schematic grid. The other three auxiliary nodes must be anchored precisely at the geometric cube roots of the primary master node's position. The electrical team will be forced to relocate all four node positions if the sum of the moduli of the positions of these four nodes is strictly less than 20 meters.

Task

Based on clear mathematical evidence for each of the three project components, provide a detailed technical advisory to the engineering firm regarding the final decisions they must make for each system.

CONSTRUCT 3: CALCULUS

ITEM 15

A local entrepreneur establishes a modern fruit processing facility in Luwero District to produce premium organic pineapple juice for the East African market.

During the pasteurization process, the extracted juice is heated to eliminate pathogens and then immediately transferred to a specialized cooling unit. The temperature, T (in °C), of the juice after t minutes in the cooling unit is modeled by the exponential function: $T(t) = 22 + 68e^{-0.04t}$. To maintain the natural flavor profile and prevent the caramelization of sugars, the rate of temperature drop exactly 25 minutes after entering the cooling unit must be greater than 1.0°C per minute. If the cooling rate falls below this critical threshold, the technical team must install high-capacity refrigeration compressors.

To pack the premium juice for retail, the marketing team designs a signature glass bottle. The outer profile of the bottle from its flat base at $x = 0$ to the neck at $x = 20$ cm is formed by rotating a boundary curve through 360° about the x-axis. The curve is modeled by the function: $y = \sqrt{0.5x + 4}$ To meet regional packaging standards for a medium-sized beverage, the total internal volume of the designed bottle must be at least 550 ml (where 1 cm^3 = 1 ml). If the volume of the calculated design is less than this standard threshold, the production team will be required to completely redesign the manufacturing mold.

In the formulation section, a large blending tank initially contains 100 liters of pure water. A concentrated liquid sugar syrup containing 2 kg of dissolved sugar per liter is pumped into the tank at a constant rate of 5 liters per minute. The well-stirred mixture continuously leaves the tank at a slower rate of 3 liters per minute. To safeguard consumer health and comply with national food safety standards, the concentration of sugar in the tank must not exceed 1.5 kg/L at the end of a 20-minute blending cycle. If the sugar concentration is projected to exceed this safety limit, the quality control manager must activate an automatic water-dilution valve.

Task:

Basing on Mathematical calculations guide the factory management to make valid decisions.

ITEM 16

A civil engineering company has been contracted to design and monitor a large cylindrical water storage tank for an irrigation project serving several nearby farmlands. The water tank is made from a strong and durable thermal plastic. During normal operations, the amount of water changes continuously as the water is pumped into the tank and released through a network of irrigation pipes to different farmlands. As the water level changes, the pressure exerted on the walls of the tank also changes. This causes small expansions that may affect the tank's dimensions. The engineers therefore need to use differential calculus to understand how the small changes in the radius may affect the volume and the rate at which the volume of the water changes.

The engineer measures the radius of the tank as 3 m and height as 10 m. The inlet pipe pours water into the tank and the height changes at a rate of 0.025 m per minute as recorded from its monitoring system. The pipe started pouring water at $6:00am$. The engineering team wishes that by 1:00 pm, the tank is full so they can start supplying the farmlands with irrigation outlet pipes.

On a certain sunny day, the engineering team measures the radius of the tank as $(244)^{1/5}$ displayed by the new calibrating machine. The engineer wishes to know the exact radius of the tank using the calculus of small changes. The machine will be approved for use if and only if the radius is $3\frac{1}{405}$ m.

One outlet pipe pours water mixed with chemical fertilizer to the nearby farmland. The engineering team notices that the path followed by the water from the pipe to the field is modelled by the differential equation $3x^2\frac{dy}{dx} = y^2$, where x is the horizontal distance covered and y is the height reached by the water. The team requires a substitution of $y = Ux$ to determine the vertical distance moved when the distance is 12 m as this ensures maximum pressure. The monitoring system records $y(2) = 6$.

Task: Help the engineering team to

a) Know whether the tank will be full by 1:00 pm.

b) Determine whether the calibration machine will be approved.

c) Determine the height of the water that ensures maximum pressure.

ITEM 17

A community in Mityana District plans to construct a water-storage facility. Before construction begins, the project engineer asks a trainee technician to assess the proposed design.

The profile of a curved section of the proposed storage chamber is represented by $y = \frac{1}{x(x-1)}$, on the coordinate grid where x and y are measured in metres. The section of the profile between $x = 2$ m and $x = 3$ m is to rotated about the x-axis through 2π radians. The project requires this section of the chamber to have a capacity of at least $0.20\,\text{m}^3$ so as to be retained or redesigned.

One of the supporting boundaries of the storage facility is represented by $x^2 - x\cos y = 1$. A supporting straight pillar is to be attached on the boundary at $\left(1, \frac{\pi}{2}\right)$.For safe attachment, the magnitude of the extent to which the pillar is sloping at this point must not exceed 1.5 otherwise the pillar is unsafe to be attached.

During filling, the relationship between the water level y and time t in seconds is modelled by $t^2\frac{dy}{dt} = t^2 + y^2$. when the time is $2s$, the water level is $1m$. Its anticipated that the storage facility must have a water level of at least 1.8 m when $3s$ have elapsed.If this is not met, the inflow rate must be increased.

Task

Using mathematical justifications, help the Mityana community make proper decisions.

ITEM 18

A food-processing company in Luzira industrial produces packaged food products. The company is on the course of developing a new package and cooling system.

The company wants to manufacture an open cardboard box with a square base of volume $1000\,\text{cm}^3$. The company wants to use the least possible amount of packaging material. The company budget allows $500\,\text{cm}^2$ of material for each box .Anything less, the company packs less and anything above, the company seeks for more funding from the head quarters.

A curved surface of the packaging machine has a cross-sectional profile described by $f(x) = \frac{x^2}{\sqrt{x^2+4}}$,in square meters. The company will approve the machine operational if its total surface area within the interval $0 \leq x \leq 2$ is beyond $1.5m^2$

After processing, a food product is placed in a cooling room. Its temperature changes at a rate proportional to the excess temperature of the surrounding ,. The surrounding temperature is maintained at 20°C.

At the beginning of cooling, the product has a temperature of 90°C. After 10 minutes, its temperature is observed to be 60°C. The company will only proceed with the packaging only if the product's temperature is below 40°C for further 10 minutes.

Task

Using mathematics calculations, help the company make valid decisions

ITEM 19

A rural water project committee in Masaka District plans to install a conical reservoir.

The reservoir has a constant vertical angle of 60°. The measured base area has a possible error of 1.5%. The project plan allows a maximum volume error of 2% otherwise the installation may not be approved.

During filling, the relationship between time t seconds and water level y meters is $t\frac{dy}{dt} - t^5e^t = 4y$, with initial conditions stating that $y(1) = 1$

Project plans state that the reservoir must have a water level of at least 82 m after 2 seconds otherwise the project should recommend buying a new reservoir.

The acceleration$a\ m/s^2$ of a certain machine control component changes according to $a = \frac{t^2-9t+5}{(t-1)^2(t+2)}$.

Determine the change in velocity between $t = 2$ s and $t = 3$ s.

The component can safely operate if the magnitude of its velocity does not exceed 1.5 m/s otherwise it needs replacement.

Task

Using mathematical evidence, guide the committee into making proper decisions.

ITEM 20

A farmer in Mityana District plans to install a long concrete water trough for livestock. The farmer wants to determine whether the proposed water-flow rate is suitable and whether the control and pressure systems will operate effectively.

The trough is 5 m long and has an equilateral triangular cross-section of side 80 cm. Water enters the trough at a constant rate of $100\sqrt{3}\,\text{cm}^3/\text{s}$. The considers anything above 0.20 cm/s as unsafe rate at which the depth of water is increasing after the water has been flowing for 25 minutes.

Determine the rate at which the depth of water is increasing

The farmer considers a rate greater than unsafe.

The performance index of a water-control component is represented by $f(x) = \frac{\sin x}{\sin x+\cos x}$, for $0 \le x \le \frac{\pi}{4}$. The required performance index is obtained from the total accumulated value of $f(x)$ over this operating

interval. The component is considered satisfactory when the index is at least 0.30.

An air-filled balloon is connected to the water-control system. The rate at which air escapes from the balloon is directly proportional to the volume of air remaining. Initially the volume is $1000\,\text{cm}^3$ and air escapes at a rate of 0.08 per second. The pressure system requires at least $500\,\text{cm}^3$ of air after 10 seconds for sustainability purposes.

Task

Using the mathematics justifications, help the farmer make proper decisions.

ITEM 21

A company in Kampala manufactures conical containers for packaging agricultural products. The company wants to maximise the capacity of the containers while controlling the shape and amount of material used.

A small right circular cone is to be fit perfectly inside a larger cone of radius 8 cm and height 18 cm, such thta the circular base of the smaller cone faces the vertex of a larger cone. If the maximum volume is less than $200\,\text{cm}^3$, the technician should recommend increasing the dimensions of the outer cone. If it is $200\,\text{cm}^3$ or more, the proposed container design should be retained.

A point on a curved surface of the container is represented parametrically on a coordinate grid by $x = \ln(1-t), \qquad y = \ln(1-t^2)$.

The acceptable curvature index at this point $\frac{1}{2}$ should satisfy $\left|\frac{d^2y}{dx^2}\right| \leq 2$. otherwise the technician should recommend modifying the surface design

The material function for a curved section of the container is $f(x) = x\sin^2 x$ over the interval $0 \leq x \leq \frac{\pi}{2}$. The company allows a maximum material index of 0.40. If the calculated index exceeds 0.40, the technician should recommend redesigning the curved section to reduce material use

Task

Basing on mathematical justifications, help the company make valid decisions in its operations.